Youssef Bazzaoui

Data Scientist | AI Consultant

bazzaoui.youssef@gmail.com +33695759559 44, avenue de Flandre, Paris 1994-05-16

youssef.bazzaoui.com linkedin.com/in/youssef-bazzaoui github.com/youssefbaz

PROFILE

Data Scientist with dual Master's degrees in Data Science and Artificial Intelligence. Skilled in predictive modeling, time series forecasting, Bayesian methods, and intelligent systems. Proven experience delivering AI-driven solutions for pricing, demand forecasting, and document analysis. Strong ability to design end-to-end data solutions, from data processing and modeling to deployment, with a focus on clarity, robustness, and decision support. Keen to apply data science and AI in consulting environments to drive actionable insights and strategic impact.

PROFESSIONAL EXPERIENCE

Freelance Data Scientist 06/2023 – Present
Data Insights (my own company) Paris, France
Designed modular AI systems for pricing, recruitment, environmental sensing with a focus on robustness, clarity, and deployment under uncertainty.

- **AI Research Assistant (LangChain/OpenAI)** Developed a modular assistant for extracting and structuring insights from academic documents, leveraging LangChain and vector-based retrieval (RAG). Focused on scalable architecture and logical information flow.
- **Forecasting & Pricing System - Siyalux (Luxembourg)** Built a dynamic pricing engine using time series forecasting and elasticity-based logic. Optimized for data sparsity, input volatility, and adaptive calibration under shifting market conditions.
- **Semantic Matching Tool for Recruitment** Created a lightweight job/profile matching system using retrieval augmented generation. Designed for responsiveness, logical transparency, and user trust through clean interface behavior.

Additional projects available upon request, including hotel pricing optimization and agent-based simulations for distributed coordination and adaptive system behavior.

Data Science Intern 12/2022 – 05/2023
Luxembourg Eurobank Private Bank Luxembourg

- Applied machine learning techniques to develop a predictive model estimating the number of products by type (Foreign Exchanges, Securities Trades, Loans, and Deposits).
- Designed and implemented SSIS packages to load data from the core banking system into the data warehouse. Developed Power BI dashboards tailored to business needs and consolidated insights across multiple data sources.

Junior Developer 09/2020 – 09/2021
Valsesia Casablanca, Morocco

- Assisted in the development and maintenance of the company's internal stock and order tracking system using Python and Excel automation. Created small scripts to automate repetitive tasks such as daily sales reporting and data entry. Gained hands-on experience in debugging, testing, and improving existing code.

EDUCATION

Master's Degree in Computer Science, Specialization in Artificial Intelligence 07/2020
University of Luxembourg Esch-sur-Alzette, Luxembourg
Relevant coursework and projects: Deep learning, probabilistic modeling, signal processing, simulation.

Bachelor's Degree in Mathematics and Computer Science Applied to Social Sciences (MIAGE) 07/2017
Université Paris Nanterre Paris, France

Applied MSc in Data Science and Artificial Intelligence 03/2023
DataScience Tech Institute Paris, France

SELECTED TECHNICAL PROJECTS

Master's Thesis (University of Luxembourg)
Machine Learning-Based Situation Awareness for Autonomous Drones
Created a real-time control system for multirotor drones based on monocular vision. Integrated safety logic, adaptive decision making, and perception under uncertainty. Demonstrated modular system design in dynamic environments, principles that align with navigation, reliability, and responsiveness in embedded intelligent systems.

Probabilistic Modeling of Fatigue Failure (with SAFRAN)
Implemented a Bayesian weakest-link model to estimate failure probabilities of turbine components under low cycle fatigue. Applied MCMC sampling for parameter inference using sparse experimental data. Contributed to the damage model design and built a Streamlit interface for visualizing failure surfaces based on material stress conditions. Demonstrated adaptability to unfamiliar scientific concepts and cross domain collaboration with industry experts.

Time Series Modeling & System Calibration
Designed forecasting and elasticity driven optimization models for pricing strategies in data sparse contexts. Emphasized robustness, feedback based adaptation, and signal level calibration under uncertainty, a foundation for thinking about correctness and resilience in real systems.

AREAS OF EXPERTISE

Deep Learning & Time Series Modeling: Time Series Forecasting Neural Networks Representation Learning Elasticity-Based Models Adaptive Calibration. Signal Processing

Behavioral Modeling & Simulation: Multi-Agent Systems Rule-Based Behavior Simulation Logic Self-Supervised Learning. Data Sparsity Handling - Modular Decision Systems

Programming & Tools: Python C++ PyTorch SQL Pandas NumPy Streamlit Git NetLogo LangChain (project-based)

SKILLS

Python | C++ | SQL | Git | Pandas | Numpy | Streamlit | NetLogo

CERTIFICATES

Bayesian Statistics using R
(edX-University of Canterburry- in progress)

Supervised Machine Learning Regression and Classification
(Coursera- Stanford University)

Python for Data Science and Machine Learning Bootcamp
(Udemy)

Amazon AWS Cloud Computing
(In Progress)

Power BI A-Z: Hands-On Power BI Training for Data Science
(Udemy)

Neo4j Certified Professional

LANGUAGES

English • **French** • **Spanish** — (basic) • **Arabic**

INTERESTS

Football | Esports | Travel & Cross-Cultural Exploration